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Latent functional Gaussian process incorporating output spatial correlations

Ziyu Liu[a], Yongxiang Li[a], Xiaowei Yue[b], and Ershun Pan[a]

[a]Department of Industrial Engineering and Management, Shanghai Jiao Tong University, Shanghai, China; [b]Department of Industrial Engineering, Tsinghua University, Beijing, China

ABSTRACT

Expensive physical experiments and time-consuming computer simulations with functional responses have spurred the development of functional surrogate models, which are designed to replace these computer simulations with statistical models. However, conventional functional surrogate modeling often relies on two-stage approaches, which often trigger information loss and fail to adequately capture output spatial correlations. This study proposes a novel one-stage method called Latent Functional Gaussian Process (LFGP) to establish a functional surrogate model. The main idea is to use B-spline basis functions to represent the functional responses, with the coefficients treated as random latent variables following a multivariate Gaussian process. A correlation kernel function called a latent functional B-spline kernel is proposed for LFGP to capture the output spatial correlations in the functional responses. To expedite the proposed one-stage modeling, efficient computational techniques are developed based on the matrix inverse lemma and the Sylvester determinant theorem. Numerical examples demonstrate the superior performance of LFGP over conventional methods. By incorporating closed shape constraints into LFGP, a closed-curve surrogate model is developed and applied to predict dimensional deformations in composite fuselage shape control.

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Functional surrogate modeling; computer experiments; blind Kriging; Gaussian process; fuselage shape control

1. Introduction

Functional responses are common in engineering practice, such as mechanical property curves of materials in additive manufacturing processes (Saunders *et al*., 2023), the multichannel profile detection in Raman spectroscopy (Yue *et al*., 2017), hydro-meteorological spatiotemporal curves (López-Lopera *et al*., 2022), and dimensional deformations in fuselage assembly (Yue *et al*., 2020). The complex interactions between system inputs and functional responses lead to expensive physical experiments and time-consuming computer simulations (Jiang *et al*., 2021). Therefore, it is essential to introduce a cost-effective and efficient alternative to predict the behavior of complex systems with functional responses.

The initial study on functional surrogate modeling aimed to develop Gaussian Process Functional Regression (GPFR) models by including functional parameters as additional inputs to the surrogate models (Shi *et al*., 2007; Liu and West, 2009). These GPFR models may face the curse of dimensionality and mismatched scales of kernel parameters (Joseph and Kang, 2011). To overcome the limitations of these models, modeling approaches based on separable covariance matrices have been introduced (Joseph *et al*., 2008; Rougier, 2008; Bayarri *et al*., 2009). Liu *et al*. (2019) developed a multiple-response Gaussian process model with a separable covariance function to estimate sensitivity indices for functional responses. Hung *et al*. (2015) proposed an expectation-maximization algorithm to construct regular grids for the Kronecker product-based functional surrogate modeling. However, these modeling approaches are limited by the regular grid condition and can only predict the functional responses at the same locations.

In addition to the construction of separable covariance matrices, considerable efforts have been made to reduce the dimensionality of functional responses, which is crucial because high-dimensional outputs can be difficult to deal with efficiently. Banerjee *et al*. (2008) and Del Castillo *et al*. (2012) used specific design matrices to represent the random effects of GPFR models in a low-dimensional subspace. Lawrence and Hyvärinen (2005) and Zhang *et al*. (2020) proposed a Gaussian process latent variable model to analyze functional responses. Perrin (2020) applied a reduced basis approach to the parameter calibration of adaptive functional Gaussian processes. These investigations mainly focused on output dimensionality reduction to facilitate Multivariate Gaussian Process (MGP) modeling and parameter calibration.

In the context of functional surrogate modeling, two-stage methods have recently been of interest due to their flexibility in representing functional responses. The first stage involves the preprocessing of functional data, where low-dimensional feature variables (basis coefficients) are extracted from functional data using some basis functions. Among these basis functions, predefined basis functions

CONTACT Yongxiang Li yongxiangli@sjtu.edu.cn
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such as Legendre polynomials (Campbell *et al.*, 2006) were constructed using consistent mathematical expressions. Adaptive basis functions were derived by fitting functional data, including, but not limited to, principal components (Higdon *et al.*, 2008; Huang and Gramacy, 2022), wavelet basis (Lu *et al.*, 2019), and Fourier basis (Konzen *et al.*, 2021). Due to the regular grid condition, these basis functions cannot be employed to represent the functional responses off the grid. Therefore, Chung and Kontar (2020) proposed functional principal components to model the functional responses over irregular grids.

In the second stage, leveraging the feature variables extracted in the first stage, a surrogate model is developed to predict the feature values of untested inputs. These predicted feature values are then used to reconstruct the functional responses. For the surrogate modeling of feature variables in this stage, most methods assumed that the feature variables are mutually independent by establishing a univariate Gaussian Process (GP) model for each variable separately (Guo *et al.*, 2017; Ma *et al.*, 2022). Due to the low dimensionality and independence of the feature variables, these GP models can be constructed rapidly in the second stage. Moreover, some joint modeling approaches have been utilized to model these feature variables employing MGP. For example, Conti and O'Hagan (2010) and Lu *et al.* (2019) developed MGP models with a separable covariance structure for these feature variables.

In summary, with the support of adaptive basis functions, two-stage methods demonstrate high computational efficiency and flexibility by constructing surrogate models for low-dimensional feature variables to represent functional responses under both regular and irregular grid conditions (Konzen *et al.*, 2021; Ma *et al.*, 2022). However, when modeling complex nonlinear functional responses, these two-stage methods may trigger the following two issues. First, projecting functional data into a low-dimensional feature space using basis functions (Fang *et al.*, 2017; Lu *et al.*, 2019; Konzen *et al.*, 2021) potentially result in the loss of functional information. Second, the feature variables used for modeling are generally assumed to be independent (Guo *et al.*, 2017; Ma *et al.*, 2022), which makes these methods inadequate to capture output spatial correlations. As an important component of functional surrogate modeling, output spatial correlations refer to the dependencies between functional responses at different locations in the global output space. Therefore, ignoring the output spatial correlations may lead to less accurate predictions.

In order to address the above issues, this study proposes a novel one-stage method for functional surrogate modeling, which is called Latent Functional Gaussian Process (LFGP) in this study. Initially, we use the linear combination of the B-spline basis to represent the functional responses. The control points (coefficients) of the B-spline basis are treated as random latent variables following MGP. Leveraging this representation, a kernel function called Latent Functional B-Spline (LFBS) kernel is proposed to capture the output spatial correlations. Subsequently, we develop LFGP based on this kernel function to simultaneously represent responses at arbitrary locations on the function. Within the proposed framework, the raw (original) functional data can be better utilized to estimate both model parameters and latent variables, facilitating accurate predictions of functional responses. It should be noted that the raw functional data refers to the original functional responses collected on regular or irregular grids, which is different from the processed feature data used in two-stage modeling methods.

To speed up the proposed one-stage functional modeling, accelerated computational techniques for LFGP under both irregular and regular grid conditions are investigated. Under the irregular grid condition, the matrix inverse lemma (Sherman and Morrison, 1950) and the Sylvester determinant theorem (Akritas *et al.*, 1996) are used to reduce the computational complexity of correlation matrix inversion and determinant solving, respectively, which accelerates the parameter estimation and prediction procedure of LFGP. Under the regular grid condition, the Kronecker product (Stegle *et al.*, 2011) is used to relax the complexity of correlation matrix inversion and determinant, according to the characteristics of regular grids.

By incorporating closed shape constraints, the proposed LFGP model is extended to the closed-shape surrogate modeling. Recently, Luo and Strait (2022) proposed MGP with a multilevel periodic kernel to predict the two-dimensional Euclidean coordinates of closed curves. This model can capture the dependencies and structural similarities of closed curves, making it effective for representing complex planar closed shapes. However, this method does not consider the influence of external factors on the closed shapes. While LFGP takes the external factors as inputs and the closed shape curves as outputs, it models the functional relationships to predict the closed shapes under varying external factors. Therefore, LFGP can be applied, for example, to predict the composite fuselage shape in aircraft assembly (Wen *et al.*, 2018). In this assembly, 10 actuators are evenly distributed around the edge of the lower half-circle of the fuselage, exerting forces to adjust the composite fuselage shape. This adjustment process is performed by experienced engineers and is typically time-consuming, which calls for advanced surrogate modeling techniques. In the case study, LFGP can more accurately model the shape dimensions of the composite fuselage due to its ability to capture the output spatial correlations and to represent the functional response at arbitrary locations.

Conventional functional surrogate models (Higdon *et al.*, 2008; Rougier, 2008; Liu *et al.*, 2019; Konzen *et al.*, 2021) often have limitations in predicting irregular-grid functional responses due to the regular grid assumption. In contrast, the proposed LFGP model can work effectively with irregular grids, providing flexibility in predicting functional responses at arbitrary locations. Unlike surrogate models using separable correlation functions such as blind Kriging (Joseph *et al.*, 2008), which induce a kind of Markov property (O'Hagan, 1998; Fricker *et al.*, 2013) that hinders the correlation information sharing across functional responses, the LFBS kernel proposed in LFGP is nonseparable, which improves the transfer of correlation information between functional responses at different locations. Unlike the two-stage methods that often involve extracting basis functions and eigenvalues from functional data (Campbell *et al.*, 2006; Fang *et al.*, 2017; Lu *et al.*, 2019), LFGP does not require these intermediate steps. Instead, the direct use of raw functional data in LFGP reduces the loss of functional information and improves the accuracy of functional modeling.

It is worth noting that in most two-stage methods, both basis functions and feature variables are typically assumed to be independent (Higdon *et al.*, 2008; Guo *et al.*, 2017). In contrast, as discussed in Section 3.1, the B-spline basis is constructed by recursively interpolating lower-order bases, which introduces spatial correlations among adjacent knots of the B-spline. Therefore, the output spatial correlations in LFGP can be jointly represented by the spatial correlations of adjacent B-spline knots and the control points. Specifically, the spatial correlations of adjacent knots capture the local dependencies of functional responses, and then the control points that follow MGP link these dependencies to represent the global dependencies of functional responses. If control points are assumed to be independent, such as the LFGP model under the independent coefficient assumption (LFGP-IC) proposed in Section 5, the model expressiveness would be limited because only the local dependencies of functional responses would be captured, affecting the model prediction accuracy.

The remainder of this article is organized as follows. Section 2 illustrates the literature review on two-stage methods for functional surrogate modeling. Section 3 introduces the construction of the LFBS kernel and LFGP modeling for both general functional responses and closed curves. The efficient computational techniques for LFGP under irregular and regular grid conditions are investigated in Section 4. The performance of LFGP over conventional methods is assessed by two synthetic examples in Section 5. A real case on composite fuselage shape control in Section 6 further validates the superiority of LFGP. Finally, the conclusions of this study and future works are presented in Section 7. The MATLAB code for this work is available on GitHub[1].

2. Literature review

In this section, we review the relevant literature in the field of functional surrogate modeling, which aims to represent the intricate relationships between system inputs and functional responses. Given a set of n inputs $\boldsymbol{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$, each input corresponds to m functional responses at the output locations $\boldsymbol{\omega} = \{\omega_1, \dots, \omega_m\}$. A foundational surrogate model (Joseph *et al.*, 2008; Liu and West, 2009) for these functional responses is

$$y(\boldsymbol{x}, \omega) = f(\boldsymbol{x}, \omega) + \epsilon(\boldsymbol{x}, \omega),$$

where $y(\boldsymbol{x}, \omega)$ is the functional response at an output location $\omega \in \mathbb{R}$ for a given input $\boldsymbol{x} \in \mathbb{R}^p$, $\epsilon(\boldsymbol{x}, \omega)$ denotes the measurement error. It is worth noting that we assume that all the input variables $\boldsymbol{x}$ are quantitative in this study, i.e., the proposed LFGP model considers only quantitative factors.

Modeling the functional response $f(\boldsymbol{x}, \omega)$ using GP may be challenging when applied to high-dimensional functional data. To address this challenge and capture the intricate relationship between the input $\boldsymbol{x}$ and the functional response $y(\boldsymbol{x}, \omega)$, some two-stage modeling methods (Higdon *et al.*, 2008; Guo *et al.*, 2017; Lu *et al.*, 2019; Ma *et al.*, 2022) have emerged as common surrogate models for functional responses. These methods provide dimensionality reduction techniques to solve the inherent complexity of high-dimensional functional data (Lawrence and Hyvärinen, 2005; Zhang *et al.*, 2020), enabling more effective modeling of the relationship between system inputs and functional responses.

In the first stage, the single functional response $y(\boldsymbol{x}, \omega)$ is discretely represented using basis functions and corresponding feature variables. This representation can be written as

$$y(\boldsymbol{x}, \omega) = \boldsymbol{b}(\omega)^T \boldsymbol{v}(\boldsymbol{x}), \tag{1}$$

where $\boldsymbol{v}(\boldsymbol{x}) = [v_1(\boldsymbol{x}), \dots, v_k(\boldsymbol{x})]^T \in \mathbb{R}^k$ denotes the hidden feature variables of functional responses for the input $\boldsymbol{x}$, and $\boldsymbol{b}(\omega) = [b_1(\omega), b_2(\omega), \dots, b_k(\omega)]^T$ represents k basis functions of output location ω. Various types of basis functions have been employed to construct $\boldsymbol{b}(\omega)$, including predefined basis (e.g., Legendre polynomials, see Campbell *et al.* (2006)) and adaptive basis (e.g., B-spline basis and principal components, see Ghosh *et al.*, 2021; and Huang and Gramacy (2022)). However, the predefined basis functions, which are constructed by consistent mathematical expressions, may not be applicable to some real applications. While, the adaptive basis functions are often limited by the requirement of regular grids. As a result, functional principal components (Fang *et al.*, 2017; Ma *et al.*, 2022) have been proposed to represent functional responses on irregular grids. The feature variables $\boldsymbol{v}(\boldsymbol{x}_i)$ for each input $\boldsymbol{x}_i$, $i = 1, \dots, n$ can be calculated using least squares to solve the equation

$$\boldsymbol{y}(\boldsymbol{x}_i) = \boldsymbol{B}(\boldsymbol{\omega}) \boldsymbol{v}(\boldsymbol{x}_i), \tag{2}$$

where $\boldsymbol{B}(\boldsymbol{\omega}) = [\boldsymbol{b}(\omega_1), \dots, \boldsymbol{b}(\omega_m)]^T \in \mathbb{R}^{m \times k}$ and $\boldsymbol{y}(\boldsymbol{x}_i) = [y(\boldsymbol{x}_i, \omega_1), \dots, y(\boldsymbol{x}_i, \omega_m)]^T$.

In the second stage, a surrogate model is developed to represent the relationships between system inputs and feature variables, that is

$$\boldsymbol{v}(\boldsymbol{x}) = \boldsymbol{\mu}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x}) + \boldsymbol{\varepsilon}(\boldsymbol{x}), \tag{3}$$

where $\boldsymbol{\mu}(\boldsymbol{x}) = [\mu_1(\boldsymbol{x}), \dots, \mu_k(\boldsymbol{x})]^T$ represents the mean function, $\boldsymbol{\varepsilon}(\boldsymbol{x}) \in \mathbb{R}^k$ is a noise vector, $\boldsymbol{g}(\boldsymbol{x}) = [g_1(\boldsymbol{x}), \dots, g_k(\boldsymbol{x})]^T$ follows MGP. For computational convenience, separable covariance functions (Gelfand *et al.*, 2004; Higdon *et al.*, 2008) rather than nonseparable ones (Majumdar and Gelfand, 2007; Fricker *et al.*, 2013; Li and Zhou, 2016; Li *et al.*, 2018) are used in MGP. The n-order matrix $\boldsymbol{R} = \{\phi(\boldsymbol{x}_i, \boldsymbol{x}_j)\}_{i,j=1,\dots,n}$ is used to denote the correlation matrix of the inputs $\boldsymbol{X}$, which is obtained by the square exponential kernel

[1]https://github.com/Yongxiang-Li/LFGP

$$\phi(\boldsymbol{x}, \boldsymbol{x}') = \exp\left(-\sum_{i=1}^{p} \theta_i (x_i - x_i')^2\right), \tag{4}$$

where $\{\theta_1, ..., \theta_p\}$ are the correlation parameters of $\phi(\boldsymbol{x}, \boldsymbol{x}')$. The covariance matrix of the MGP model is denoted by $\sigma^2 \boldsymbol{R} \otimes \boldsymbol{\Sigma}$, where σ^2 is a variance parameter and $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$ is a cross correlation matrix. The matrix $\boldsymbol{\Sigma}$ can be either a parameterized correlation matrix (Lu *et al.*, 2019) or an identity matrix (Guo *et al.*, 2017; Ma *et al.*, 2022). Using the identity matrix for $\boldsymbol{\Sigma}$ is essentially equivalent to building a univariate GP model for each feature variable separately.

Once the feature variables $\hat{\boldsymbol{v}}(\boldsymbol{x}^*)$ for the input $\boldsymbol{x}^*$ are predicted by (3) with n sets of feature values from (2), the corresponding functional responses at locations $\boldsymbol{\omega}$ are given by

$$\hat{\boldsymbol{y}}(\boldsymbol{x}^*) = \boldsymbol{B}(\boldsymbol{\omega}) \hat{\boldsymbol{v}}(\boldsymbol{x}^*).$$

However, two-stage methods require preprocessing the functional data into the low-dimensional feature data in the first stage, and this preprocessing may not capture the full information of the functional data. In addition, the assumption of independent feature variables in two-stage methods makes it challenging to effectively represent output spatial correlations, thus affecting the performance of functional surrogate modeling. To overcome these limitations and address these issues, we propose LFGP in Section 3.

3. LFGP

In this section, we provide a detailed introduction to the modeling of LFGP. Within this modeling framework, we utilize B-spline basis to represent the functional responses, with the basis coefficients treated as random latent variables following MGP. Based on this representation, we introduce the LFBS kernel and proceed to develop the LFGP model. The proposed LFGP model can leverage raw functional data to estimate both parameters and random latent variables, allowing effective prediction of functional responses at arbitrary locations.

3.1. LFBS kernel and surrogate modeling

We begin by introducing the use of a B-spline as a fundamental component of the LFGP modeling framework. The B-spline, first proposed by De Boor and De Boor (1978), provides a versatile interpolation technique with several advantageous properties for representing complex functional responses. Specifically, the B-spline basis is generated through a recursive process, which can be described as follows

$$b_{r,0}(\omega) = \begin{cases} 1 & u_r \le \omega < u_{r+1} \\ 0 & \text{otherwise} \end{cases}$$

$$b_{r,d}(\omega) = \frac{\omega - u_r}{u_{r+d} - u_r} b_{r,d-1}(\omega) + \frac{u_{r+d+1} - \omega}{u_{r+d+1} - u_{r+1}} b_{r+1,d-1}(\omega), \tag{5}$$

for $r = 1, ..., k$, where d is the order of the B-spline basis, ω denotes the output location for interpolation, and $\{u_1, ..., u_{k+1}\}$ are knots defining the support intervals of the B-spline basis.

In (5), the higher-order basis $b_{r,d}(\omega)$ relies on the interpolation of lower-order basis $b_{r,d-1}(\omega)$ and $b_{r+1,d-1}(\omega)$. This recursive property brings spatial correlations to adjacent knots of the B-spline, as the higher-order bases corresponding to adjacent knots share common lower-order bases. These spatial correlations are used to describe the local dependencies between adjacent B-spline knots, which is a key component for representing the output spatial correlations. Furthermore, each B-spline basis can only be used to represent the functional responses within its local support interval. This local support property enables LFGP to capture local variations among functions that exhibit similar trends (Zhang *et al.*, 2017).

Suppose that the functional responses $\boldsymbol{y}_i = [y(\boldsymbol{x}_i, \omega_{i,1}), ..., y(\boldsymbol{x}_i, \omega_{i,m})]^T$ for each input $\boldsymbol{x}_i$ are collected at different output locations $\boldsymbol{\omega}_i = \{\omega_{i,1}, ..., \omega_{i,m}\}$ for $i = 1, ..., n$, where $m \le n$. Then, using the B-spline basis, the functional response $y(\boldsymbol{x}, \omega)$ can be represented as follows

$$y(\boldsymbol{x}, \omega) = \boldsymbol{f}(\boldsymbol{x})^T \boldsymbol{\beta} + \boldsymbol{b}(\omega)^T \boldsymbol{z}(\boldsymbol{x}) + \epsilon(\omega), \tag{6}$$

where $\boldsymbol{f}(\boldsymbol{x}) = [f_1(\boldsymbol{x}), ..., f_q(\boldsymbol{x})]^T \in \mathbb{R}^q$ is the regression function, $\boldsymbol{\beta} = [\beta_1, \beta_2, ..., \beta_q]^T \in \mathbb{R}^q$ is the regression parameter, $\boldsymbol{b}(\omega)$ consists of k d-order B-spline basis functions, and $\epsilon(\omega) \sim \mathcal{N}(0, \sigma^2 \delta^2)$ is an i.i.d. measurement error term.

The basis coefficients $\boldsymbol{z}(\boldsymbol{x}) = [z_1(\boldsymbol{x}), ..., z_k(\boldsymbol{x})]^T \in \mathbb{R}^k$ in (6) actually refer to the control points of the B-spline, which are spatially correlated because the functional responses they represent are spatially correlated. Therefore, $\boldsymbol{z}(\boldsymbol{x})$ are treated as latent variables to account for the underlying randomness of the functional response $y(\boldsymbol{x}, \omega)$, following a zero-mean MGP with a separable covariance function

$$\mathrm{Cov}(z_r(\boldsymbol{x}), z_t(\boldsymbol{x}')) = \sigma^2 \phi_0(r, t) \phi(\boldsymbol{x}, \boldsymbol{x}'), \quad r, t = 1, ..., k. \tag{7}$$

The correlation function in (7) is determined by two parts. $\phi(\boldsymbol{x}, \boldsymbol{x}')$ represents the correlation between different inputs $\boldsymbol{x}$ and $\boldsymbol{x}'$, while $\phi_0(r, t)$ depicts the correlation between $z_r(\boldsymbol{x})$ and $z_t(\boldsymbol{x})$. $\phi(\boldsymbol{x}, \boldsymbol{x}')$ is as presented in (4) and $\phi_0(r, t)$ is often constructed by the square exponential kernel

$$\phi_0(r, t) = \exp\left(-\theta_0 \left(\frac{r - t}{k - 1}\right)^2\right),$$

where θ_0 is the correlation parameter of $\phi_0(r, t)$. In cases where the system outputs are closed curves, such as the dimensional deformations of the composite fuselage in Section 6, the periodic kernel (MacKay *et al.*, 1998) is used for $\phi_0(r, t)$ to better characterize the closed nature of the functional responses, that is

$$\phi_0(r, t) = \exp\left(-\theta_0 \sin^2\left(\frac{\pi(r - t)}{k - 1}\right)\right).$$

In this study, we assume that the arrangement of $\boldsymbol{z}(\boldsymbol{x})$ in space remains consistent for each input $\boldsymbol{x}$. Thus, the distribution of the latent variables $\boldsymbol{z} = [\boldsymbol{z}(\boldsymbol{x}_1)^T, ..., \boldsymbol{z}(\boldsymbol{x}_n)^T]^T \in \mathbb{R}^{nk}$ for inputs $\boldsymbol{X}$ is $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{R} \otimes \boldsymbol{\Sigma}_z)$, where $\sigma^2 \boldsymbol{R} \otimes \boldsymbol{\Sigma}_z$ is a separable covariance matrix based on the Kronecker product, and $\boldsymbol{\Sigma}_{\boldsymbol{z}} = \{\phi_0(r,t)\}_{r,t=1,\dots,k}$ is the k-order matrix.

Based on the functional representation in (6) and the correlation structure of latent variables $\boldsymbol{z}$, a nonseparable kernel called the LFBS kernel is introduced to represent the output spatial correlation of functional responses, which is defined as

$$\psi_{\boldsymbol{\theta}}\big(y(\boldsymbol{x},\omega), y(\boldsymbol{x}',\omega')\big) = \phi(\boldsymbol{x},\boldsymbol{x}')\boldsymbol{b}(\omega)^T\boldsymbol{\Sigma}_{\boldsymbol{z}}\boldsymbol{b}(\omega') + \delta^2 I_{\omega=\omega'}, \tag{8}$$

where $\boldsymbol{\theta} = \{\theta_0,\theta_1,\dots,\theta_p\}$ is the set of correlation parameters of $\phi_0(r,t)$ and $\phi(\boldsymbol{x},\boldsymbol{x}')$, ω and ω' denote the output locations of functions corresponding to the inputs $\boldsymbol{x}$ and $\boldsymbol{x}'$, respectively, δ^2 is the correlation constant for measurement error, and the indicator function $I_{\omega=\omega'}$ equals 1 if $\omega=\omega'$, otherwise it equals zero.

The LFBS kernel can effectively capture the correlation of responses at arbitrary locations in the output space, which makes the proposed model suitable for cases involving functional responses on irregular grids. Proposition 1 demonstrates that the correlation matrix of functional responses based on the LFBS kernel is positive definite. The proof of Proposition 1 is given in the Appendix. Furthermore, the separable correlation structure of latent variables in this LFBS kernel allows the efficient computation of LFGP, which is discussed in detail in Section 4.2.

Proposition 1. *For any n inputs $\boldsymbol{x}_1,\boldsymbol{x}_2,\dots,\boldsymbol{x}_n$ and the corresponding functional responses $\boldsymbol{y} = [\boldsymbol{y}_1^T,\dots,\boldsymbol{y}_n^T]^T$, the correlation matrix of $\boldsymbol{y}$ using the LFBS kernel $\psi_{\boldsymbol{\theta}}(y(\boldsymbol{x},\omega), y(\boldsymbol{x}',\omega'))$ is always positive definite.*

To simultaneously represent the functional responses at arbitrary output locations, we introduce LFGP as an extension of (6), where the correlations among these functional responses are represented by the LFBS kernel in (8). In some practical situations, functional data is collected with a limited number of sensors, which requires randomization of sensor locations for repeated data collection to ensure data richness. Thus, we assume that the functional responses for each input $\boldsymbol{x}_i \in \boldsymbol{X}$ are collected s times. $\boldsymbol{y}_i^\iota = [y(\boldsymbol{x}_i,\omega_{i,1}^\iota),\dots,y(\boldsymbol{x}_i,\omega_{i,m}^\iota)]^T \in \mathbb{R}^m$ are the functional responses at locations $\boldsymbol{\omega}_i^\iota = \{\omega_{i,1}^\iota,\dots,\omega_{i,m}^\iota\}$ of the ιth collection, for $\iota = 1,\dots,s$. Then, the LFGP for functional responses $\boldsymbol{Y}_\iota = [(\boldsymbol{y}_1^\iota)^T,\dots,(\boldsymbol{y}_n^\iota)^T]^T \in \mathbb{R}^{mn}$ can be written as

$$\boldsymbol{Y}_\iota = \boldsymbol{f}\boldsymbol{\beta} + \boldsymbol{B}_\iota\boldsymbol{z} + \boldsymbol{\epsilon}_\iota,$$

where $\boldsymbol{f} = [\boldsymbol{f}_1^T,\dots,\boldsymbol{f}_n^T]^T$, $\boldsymbol{f}_i = \mathbf{1}_m \otimes \boldsymbol{f}(\boldsymbol{x}_i)^T$, $\mathbf{1}_m$ is an $m\times 1$ vector filled with 1, $\boldsymbol{B}_\iota = \mathrm{diag}[\boldsymbol{B}(\boldsymbol{\omega}_1^\iota),\dots,\boldsymbol{B}(\boldsymbol{\omega}_n^\iota)] \in \mathbb{R}^{mn\times nk}$, $\boldsymbol{B}(\boldsymbol{\omega}_i^\iota) = [\boldsymbol{b}(\omega_{i,1}^\iota),\dots,\boldsymbol{b}(\omega_{i,m}^\iota)]^T \in \mathbb{R}^{m\times k}$, and $\boldsymbol{\epsilon}_\iota$ is the measurement error term.

Denoting all the collected functional responses as $\boldsymbol{Y} = [\boldsymbol{Y}_1^T,\dots,\boldsymbol{Y}_s^T]^T \in \mathbb{R}^{smn}$, the LFGP model is established as follows

$$\boldsymbol{Y} = \boldsymbol{F}\boldsymbol{\beta} + \boldsymbol{B}\boldsymbol{z} + \boldsymbol{\epsilon}, \tag{9}$$

where $\boldsymbol{F} = \mathbf{1}_s \otimes \boldsymbol{f}$, $\boldsymbol{B} = [\boldsymbol{B}_1^T,\dots,\boldsymbol{B}_s^T]^T \in \mathbb{R}^{smn\times nk}$ and $\boldsymbol{\epsilon} = [\boldsymbol{\epsilon}_1^T,\dots,\boldsymbol{\epsilon}_s^T]^T$. Assuming mutual independence between the latent variables $\boldsymbol{z}$ and the measurement errors $\boldsymbol{\epsilon}$, the distribution of $\boldsymbol{Y}$ is defined as

$$\boldsymbol{Y} \sim \mathcal{N}\Big(\boldsymbol{F}\boldsymbol{\beta}, \sigma^2\boldsymbol{\Sigma}_y\Big), \tag{10}$$

where $\boldsymbol{\Sigma}_y = \boldsymbol{\Sigma}_b + \delta^2\boldsymbol{I}_{smn}$, $\boldsymbol{\Sigma}_b = \boldsymbol{B}(\boldsymbol{R}\otimes\boldsymbol{\Sigma}_{\boldsymbol{z}})\boldsymbol{B}^T$, and the elements of $\boldsymbol{\Sigma}_y$ is determined by (8). Here, $\boldsymbol{I}_{smn}$ is the smn-order identity matrix.

3.2. Parameter estimation of LFGP

In accordance with the distribution in (10), we can derive the log-likelihood function given the functional responses $\boldsymbol{Y}$ as follows

$$\ell(\boldsymbol{\theta},\boldsymbol{\beta},\sigma^2,\delta^2) = -\frac{1}{2}\left(\log|\sigma^2\boldsymbol{\Sigma}_y| + \frac{1}{\sigma^2}(\boldsymbol{Y}-\boldsymbol{F}\boldsymbol{\beta})^T\boldsymbol{\Sigma}_y^{-1}(\boldsymbol{Y}-\boldsymbol{F}\boldsymbol{\beta})\right). \tag{11}$$

Given $\{\boldsymbol{\theta},\delta^2\}$, by solving the equation where the partial derivatives of (11) with respect to $\boldsymbol{\beta}$ and σ^2 equal to zero (Myung, 2003), the estimates of $\boldsymbol{\beta}$ and σ^2 are

$$\begin{cases} \hat{\boldsymbol{\beta}} &= \left(\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{F}\right)^{-1}\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{Y} \\ \hat{\sigma}^2 &= \dfrac{1}{smn}\big(\boldsymbol{Y}-\boldsymbol{F}\hat{\boldsymbol{\beta}}\big)^T\boldsymbol{\Sigma}_y^{-1}\big(\boldsymbol{Y}-\boldsymbol{F}\hat{\boldsymbol{\beta}}\big). \end{cases} \tag{12}$$

By plugging the estimated $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ into (11), we obtain the simplified log-likelihood function

$$\ell(\boldsymbol{\theta},\delta^2) = -\frac{1}{2}\Big(smn\log\hat{\sigma}^2 + \log|\boldsymbol{\Sigma}_y|\Big). \tag{13}$$

Finally, the parameters $\hat{\boldsymbol{\theta}}$ and $\hat{\delta}^2$ are determined by maximizing (13) using optimal search algorithms.

3.3. Latent variable-based functional prediction

With the model parameters that have been estimated in Section 3.2, the fitted LFGP can be used to predict functional responses for any untested input $\boldsymbol{x}^* = [x_1^*, x_2^*,\dots,x_p^*]$. Unlike conventional two-stage methods (Guo *et al.*, 2017; Ma *et al.*, 2022) that require preprocessing of functional data to extract basis functions and eigenvalues, LFGP can work directly with the collected raw functional data $\boldsymbol{Y}$ to predict the latent variables $\boldsymbol{z}^* = [z_1(\boldsymbol{x}^*),\dots,z_k(\boldsymbol{x}^*)]^T$ associated with the input $\boldsymbol{x}^*$.

To enable this prediction, we jointly model the functional responses $\boldsymbol{Y}$ and the latent variables $\boldsymbol{z}^*$ in the LFGP framework as follows

$$\begin{bmatrix}\boldsymbol{Y}\\ \boldsymbol{z}^*\end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\boldsymbol{F}\hat{\boldsymbol{\beta}}\\ \boldsymbol{0}\end{bmatrix}, \begin{bmatrix}\hat{\sigma}^2\boldsymbol{\Sigma}_y & \hat{\sigma}^2\boldsymbol{C}_{zy}(\boldsymbol{X},\boldsymbol{x}^*)\\ \hat{\sigma}^2\boldsymbol{C}_{zy}(\boldsymbol{x}^*,\boldsymbol{X}) & \hat{\sigma}^2\boldsymbol{\Sigma}_{\boldsymbol{z}}\end{bmatrix}\right), \tag{14}$$

where the correlation matrix between $\boldsymbol{Y}$ and $\boldsymbol{z}^*$ is $\boldsymbol{C}_{zy}(\boldsymbol{X},\boldsymbol{x}^*) = \boldsymbol{B}(\boldsymbol{R}(\boldsymbol{X},\boldsymbol{x}^*)\otimes\boldsymbol{\Sigma}_{\boldsymbol{z}}) \in \mathbb{R}^{smn\times k}$.

By inferring the conditional distribution $p(\boldsymbol{z}^*|\boldsymbol{Y})$ from (14), the prediction of $\boldsymbol{z}^*$ can be represented as

$$\begin{cases} \hat{\boldsymbol{z}}^* &= \boldsymbol{C}_{zy}(\boldsymbol{x}^*,\boldsymbol{X})\boldsymbol{\Sigma}_y^{-1}\big(\boldsymbol{Y}-\boldsymbol{F}\hat{\boldsymbol{\beta}}\big) \\ \mathrm{Cov}(\hat{\boldsymbol{z}}^*) &= \hat{\sigma}^2\boldsymbol{\Sigma}_{\boldsymbol{z}} - \hat{\sigma}^2\boldsymbol{C}_{zy}(\boldsymbol{x}^*,\boldsymbol{X})\boldsymbol{\Sigma}_y^{-1}\boldsymbol{C}_{zy}(\boldsymbol{X},\boldsymbol{x}^*). \end{cases} \tag{15}$$

Thus, following (9), the prediction of $\boldsymbol{y}^* = [y(\boldsymbol{x}^*,\omega_1^*),\dots,y(\boldsymbol{x}^*,\omega_m^*)]^T$ at arbitrary output locations $\boldsymbol{\omega}^* = \{\omega_1^*,\dots,\omega_m^*\}$ can be expressed as

$$\begin{cases} \hat{\boldsymbol{y}}^* & = \boldsymbol{f}^*\hat{\boldsymbol{\beta}} + \boldsymbol{B}(\boldsymbol{\omega}^*)\hat{\boldsymbol{z}}^* \\ \mathrm{Cov}(\hat{\boldsymbol{y}}^*) & = \boldsymbol{B}(\boldsymbol{\omega}^*)\mathrm{Cov}(\hat{\boldsymbol{z}}^*)\boldsymbol{B}(\boldsymbol{\omega}^*)^T + \hat{\sigma}^2\hat{\delta}^2\boldsymbol{I}_m, \end{cases} \tag{16}$$

where $\boldsymbol{f}^* = \mathbf{1}_m \otimes \boldsymbol{f}(\boldsymbol{x}^*)^T$, $\boldsymbol{B}(\boldsymbol{\omega}^*) = \left[\boldsymbol{b}(\omega_1^*), ..., \boldsymbol{b}(\omega_m^*)\right]^T$.

3.4. Functional surrogate modeling for closed curves

In the modeling of LFGP, we assume that the system outputs are considered as general function curves. However, in some applications, such as the composite fuselage shape control described in Section 6, the output function may be a closed curve, which in our case study represents the dimensional deformations of the composite fuselage. Unlike general function curves, closed curves exhibit a smooth connection between their endpoints, which poses a challenge for modeling using general LFGP. Therefore, we introduce closed shape constraints into LFGP modeling to represent these closed curves. The constraints are designed to ensure that both the function values and the derivatives at the function endpoints are equal.

To illustrate the significance of the closed shape constraints, the LFGP predicted curves (depicted as red solid lines) for closed circular shapes are shown in Figure 1. It can be seen that the predicted curve with closed shape constraints displays a smoother connection at the endpoints than that without constraints, highlighted by a black circle in Figure 1(a). Specifically, the closed shape constraints can be expressed as $\boldsymbol{\Gamma}\hat{\boldsymbol{z}} = \mathbf{0}$, where $\hat{\boldsymbol{z}} = (\boldsymbol{R} \otimes \boldsymbol{\Sigma}_z)\boldsymbol{B}^T\boldsymbol{\Sigma}_y^{-1}(\boldsymbol{Y} - \boldsymbol{F}\boldsymbol{\beta})$ is the conditional mean of latent variables $\boldsymbol{z}$ for inputs $\boldsymbol{X}$,

$$\boldsymbol{\Gamma} = \boldsymbol{I}_n \otimes \begin{bmatrix} \left(\boldsymbol{b}(\omega_s) - \boldsymbol{b}(\omega_e)\right)^T \\ \left(\boldsymbol{b}'(\omega_s) - \boldsymbol{b}'(\omega_e)\right)^T\boldsymbol{\Pi} \end{bmatrix},$$

ω_s and ω_e represent the locations of the endpoints of the function, $\boldsymbol{b}'(\omega_s)^T\boldsymbol{\Pi}$ and $\boldsymbol{b}'(\omega_e)^T\boldsymbol{\Pi}$ denote the B-spline basis derivatives at the endpoints of the function, as referenced in Li *et al.* (2022).

When modeling LFGP for closed curves, the parameter estimation involves solving a constrained optimization problem. The objective function is the log-likelihood function from (11), and it is subject to the closed shape constraints

$$\max \ell(\boldsymbol{\theta}, \boldsymbol{\beta}, \sigma^2, \delta^2) = -\frac{1}{2}\left(\log|\sigma^2\boldsymbol{\Sigma}_y| + \frac{1}{\sigma^2}(\boldsymbol{Y} - \boldsymbol{F}\boldsymbol{\beta})^T\boldsymbol{\Sigma}_y^{-1}(\boldsymbol{Y} - \boldsymbol{F}\boldsymbol{\beta})\right) \quad s.t.\ \boldsymbol{\Gamma}\hat{\boldsymbol{z}} = \mathbf{0}, \tag{17}$$

Following the Lagrange multiplier method, this constrained optimization problem can be reformulated as an unconstrained one

$$\max \ell(\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\lambda}, \sigma^2, \delta^2) = -\frac{1}{2}\left(\log|\sigma^2\boldsymbol{\Sigma}_y| + \frac{1}{\sigma^2}(\boldsymbol{Y} - \boldsymbol{F}\boldsymbol{\beta})^T\boldsymbol{\Sigma}_y^{-1}(\boldsymbol{Y} - \boldsymbol{F}\boldsymbol{\beta})\right) + \frac{1}{\sigma^2}\boldsymbol{\lambda}^T\boldsymbol{\Gamma}\hat{\boldsymbol{z}}, \tag{18}$$

where $\boldsymbol{\lambda} \in \mathbb{R}^{2n}$ is the Lagrange parameter. Therefore, we can use maximum likelihood estimation (Myung, 2003) for (18). Given $\{\boldsymbol{\theta}, \delta^2\}$, the estimates of $\boldsymbol{\beta}$ and $\boldsymbol{\lambda}$ are

$$\begin{cases} \hat{\boldsymbol{\beta}} & = \left(\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{F}\right)^{-1}\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\left(\boldsymbol{Y} - \boldsymbol{B}(\boldsymbol{R} \otimes \boldsymbol{\Sigma}_z)\boldsymbol{\Gamma}^T\hat{\boldsymbol{\lambda}}\right) \\ \hat{\boldsymbol{\lambda}} & = \left(\boldsymbol{\Gamma}_y\boldsymbol{F}\left(\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{F}\right)^{-1}\boldsymbol{F}^T\boldsymbol{\Gamma}_y^T\right)^{-1}\boldsymbol{\Gamma}_y\left(\boldsymbol{F}\left(\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{F}\right)^{-1}\boldsymbol{F}^T\boldsymbol{\Sigma}_y^{-1}\boldsymbol{Y} - \boldsymbol{Y}\right), \end{cases} \tag{19}$$

where $\boldsymbol{\Gamma}_y = \boldsymbol{\Gamma}(\boldsymbol{R} \otimes \boldsymbol{\Sigma}_z)\boldsymbol{B}^T\boldsymbol{\Sigma}_y^{-1}$.

Given the model parameters estimated using (19) and any untested input $\boldsymbol{x}^*$, the corresponding closed-curve responses $\boldsymbol{y}_c^* = \left[y_c(\boldsymbol{x}^*, \omega_1^*), ..., y_c(\boldsymbol{x}^*, \omega_m^*)\right]^T$ can be predicted by LFGP as follows

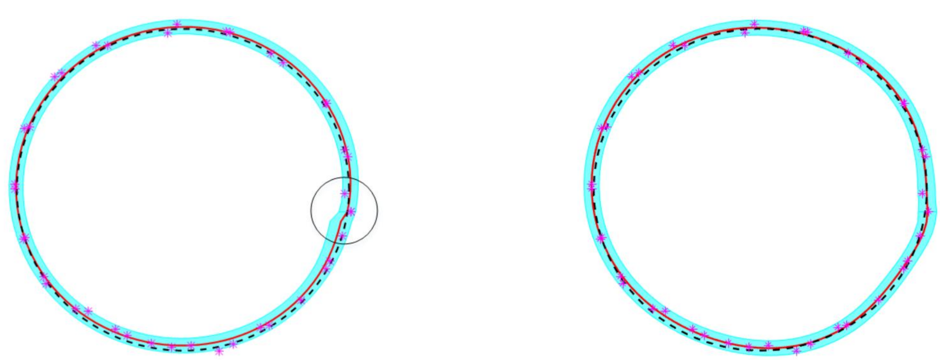
$$\begin{cases} \hat{\boldsymbol{y}}_c^* & = \boldsymbol{f}^*\hat{\boldsymbol{\beta}} + \boldsymbol{B}(\boldsymbol{\omega}^*)\hat{\boldsymbol{z}}^* \\ \mathrm{Cov}(\hat{\boldsymbol{y}}_c^*) & = \boldsymbol{B}(\boldsymbol{\omega}^*)\mathrm{Cov}(\hat{\boldsymbol{z}}^*)\boldsymbol{B}(\boldsymbol{\omega}^*)^T + \hat{\sigma}^2\hat{\delta}^2\boldsymbol{I}_m, \end{cases} \tag{20}$$

where the prediction of latent variables $\boldsymbol{z}^*$ is

$$\begin{cases} \hat{\boldsymbol{z}}^* & = \boldsymbol{C}_{zy}(\boldsymbol{x}^*, \boldsymbol{X})\boldsymbol{\Sigma}_y^{-1}\left(\boldsymbol{Y} - \boldsymbol{F}\hat{\boldsymbol{\beta}}\right) \\ \mathrm{Cov}(\hat{\boldsymbol{z}}^*) & = \hat{\sigma}^2\boldsymbol{\Sigma}_z - \hat{\sigma}^2\boldsymbol{C}_{zy}(\boldsymbol{x}^*, \boldsymbol{X})\boldsymbol{\Sigma}_y^{-1}\boldsymbol{C}_{zy}(\boldsymbol{X}, \boldsymbol{x}^*). \end{cases} \tag{21}$$

4. Efficient computation of LFGP

In this section, we investigate efficient computation of LFGP under both irregular and regular grid conditions. As a one-stage functional surrogate model that uses the original (raw)



(a) Predicted curve of LFGP without constraints (b) Predicted curve of LFGP with constraints

Figure 1. Predicted curves of LFGP for closed circular shapes.

functional data and considers the output spatial correlations, LFGP can be slow for large-scale data. Therefore, efficient computational methods are developed to circumvent the need for directly solving the inversion and determinant of the correlation matrix $\boldsymbol{\Sigma}_y$, which has a cubic complexity. It should be noted that the efficient computational techniques proposed in this section are derived for the LFGP model of general functional responses, and similar techniques are applicable to the LFGP model of closed-curve responses.

4.1. Efficient computation in irregular grid condition

In this subsection, we introduce the efficient computational techniques of LFGP under an irregular grid condition. Specifically, the irregular grid condition means that the output locations $\boldsymbol{\omega}_i = \{\omega_{i,1}, \ldots, \omega_{i,m}\}$ for each input $\boldsymbol{x}_i \in \boldsymbol{X}$ are random at each repeated collection, which is considered in the LFGP modeling in Section 3.

Thus, following the Sherman-Morrison-Woodbury formula (Sherman and Morrison, 1950) and the Sylvester determinant theorem (Akritas *et al.*, 1996), the inversion and the determinant of $\boldsymbol{\Sigma}_y$ in (10) can be written respectively as

$$\boldsymbol{\Sigma}_y^{-1} = \frac{1}{\delta^2}\boldsymbol{I}_{smn} - \frac{1}{\delta^2}\boldsymbol{B}\boldsymbol{\Omega}_y^{-1}\boldsymbol{B}^T, \tag{22}$$

$$|\boldsymbol{\Sigma}_y| = \delta^{2(smn-nk)}|\boldsymbol{R}|^k|\boldsymbol{\Sigma}_z|^n|\boldsymbol{\Omega}_y|, \tag{23}$$

where the nk-order matrix $\boldsymbol{\Omega}_y = \delta^2\boldsymbol{R}^{-1} \otimes \boldsymbol{\Sigma}_z^{-1} + \boldsymbol{B}^T\boldsymbol{B}$.

Since both $\boldsymbol{R}^{-1}$ and $\boldsymbol{\Sigma}_z^{-1}$ are positive definite, we decompose $\boldsymbol{\Omega}_y^{-1}$ into $\boldsymbol{L}\boldsymbol{L}^T$ by Cholesky factorization. Then, $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ in (12) can be written as

$$\begin{cases} \hat{\boldsymbol{\beta}} & = \left(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{F}_{\boldsymbol{B}}^T\boldsymbol{F}_{\boldsymbol{B}}\right)^{-1}\left(\boldsymbol{F}^T\boldsymbol{Y} - \boldsymbol{F}_{\boldsymbol{B}}^T\boldsymbol{Y}_{\boldsymbol{B}}\right) \\ \hat{\sigma}^2 & = \dfrac{1}{smn\delta^2}\left(||\boldsymbol{Y} - \boldsymbol{F}\hat{\boldsymbol{\beta}}||_2^2 - ||\boldsymbol{Y}_{\boldsymbol{B}} - \boldsymbol{F}_{\boldsymbol{B}}\hat{\boldsymbol{\beta}}||_2^2\right), \end{cases} \tag{24}$$

where $||\cdot||_2$ is the l_2-norm, $\boldsymbol{Y}_{\boldsymbol{B}} = (\boldsymbol{B}\boldsymbol{L})^T\boldsymbol{Y} \in \mathbb{R}^{nk}$, and $\boldsymbol{F}_{\boldsymbol{B}} = (\boldsymbol{B}\boldsymbol{L})^T\boldsymbol{F} \in \mathbb{R}^{nk\times q}$. Thus, the computational complexity of solving $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ is relaxed to $\mathcal{O}(smn(nk)^2)$.

By plugging (23) into (13), the simplified log-likelihood function is

$$\ell(\boldsymbol{\theta}, \delta^2) = -\frac{1}{2}\left(smn\log\hat{\sigma}^2 + D_y\right), \tag{25}$$

where $D_y = (smn - nk)\log\delta^2 + k\log|\boldsymbol{R}| + n\log|\boldsymbol{\Sigma}_z| + \log|\boldsymbol{\Omega}_y|$. Thus, the computational complexity of solving the log-likelihood function reduces to $\mathcal{O}(n^3k^3)$.

With the parameters estimated by (24) and (25), the prediction of latent variables $\boldsymbol{z}^*$ in (15) can be represented as

$$\begin{cases} \hat{\boldsymbol{z}}^* & = \dfrac{1}{\hat{\delta}^2}\left(\boldsymbol{C}_{zy}(\boldsymbol{x}^*, \boldsymbol{X})\boldsymbol{E}_y - \boldsymbol{C}_{\boldsymbol{B}}(\boldsymbol{x}^*, \boldsymbol{X})\boldsymbol{E}_{\boldsymbol{B}}\right) \\ \mathrm{Cov}(\hat{\boldsymbol{z}}^*) & = \hat{\sigma}^2\boldsymbol{\Sigma}_z - \dfrac{\hat{\sigma}^2}{\hat{\delta}^2}(\boldsymbol{P}_z - \boldsymbol{P}_{\boldsymbol{B}}), \end{cases} \tag{26}$$

where

$$\begin{cases} \boldsymbol{E}_y & = \boldsymbol{Y} - \boldsymbol{F}\hat{\boldsymbol{\beta}} \\ \boldsymbol{E}_{\boldsymbol{B}} & = \boldsymbol{Y}_{\boldsymbol{B}} - \boldsymbol{F}_{\boldsymbol{B}}\hat{\boldsymbol{\beta}}, \end{cases} \quad \begin{cases} \boldsymbol{P}_z & = \boldsymbol{C}_{zy}(\boldsymbol{x}^*, \boldsymbol{X})\boldsymbol{C}_{zy}(\boldsymbol{X}, \boldsymbol{x}^*) \\ \boldsymbol{P}_{\boldsymbol{B}} & = \boldsymbol{C}_{\boldsymbol{B}}(\boldsymbol{x}^*, \boldsymbol{X})\boldsymbol{C}_{\boldsymbol{B}}(\boldsymbol{X}, \boldsymbol{x}^*), \end{cases}$$

and $\boldsymbol{C}_{\boldsymbol{B}}(\boldsymbol{X}, \boldsymbol{x}^*) = (\boldsymbol{B}\boldsymbol{L})^T\boldsymbol{C}_{zy}(\boldsymbol{X}, \boldsymbol{x}^*) \in \mathbb{R}^{nk\times k}$.

In accordance with (24) and (26), it can be seen that the cubic complexity of parameter estimation and prediction is relaxed by using the matrix inverse lemma and the Sylvester determinant theorem. Consequently, the complexity of LFGP under the irregular grid condition is $\mathcal{O}(smn(nk)^2)$.

4.2. Efficient computation in regular grid condition

In this subsection, we introduce the efficient computational techniques of LFGP under the regular grid condition. Unlike the irregular grid condition with random output locations, this condition requires the functional responses for all inputs $\boldsymbol{X}$ to be collected at the same locations. Thus, given m regularly arranged output locations $\boldsymbol{\omega} = \{\omega_1, \ldots, \omega_m\}$, the B-spline basis matrix $\boldsymbol{B}$ in (9) can be expressed as a Kronecker structure $\boldsymbol{B} = \boldsymbol{V} \otimes \boldsymbol{B}(\boldsymbol{\omega})$, where $\boldsymbol{V} = \boldsymbol{1}_s \otimes \boldsymbol{I}_n$. The correlation matrix $\boldsymbol{\Sigma}_y$ of $\boldsymbol{Y}$ is represented as

$$\boldsymbol{\Sigma}_y = \boldsymbol{R}_V \otimes \boldsymbol{\Sigma}_B + \delta^2\boldsymbol{I}_{smn}, \tag{27}$$

where $\boldsymbol{R}_V = \boldsymbol{V}\boldsymbol{R}\boldsymbol{V}^T \in \mathbb{R}^{sn\times sn}$ and $\boldsymbol{\Sigma}_B = \boldsymbol{B}(\boldsymbol{\omega})\boldsymbol{\Sigma}_z\boldsymbol{B}(\boldsymbol{\omega})^T \in \mathbb{R}^{m\times m}$.

To speed up the computation of $\boldsymbol{\Sigma}_y$, we perform the eigenvalue decomposition of $\boldsymbol{R}_V$ and $\boldsymbol{\Sigma}_B$ in (27)

$$\begin{cases} \boldsymbol{R}_V & = \boldsymbol{U}_R\boldsymbol{\Lambda}_R\boldsymbol{U}_R^T \\ \boldsymbol{\Sigma}_B & = \boldsymbol{U}_\Sigma\boldsymbol{\Lambda}_\Sigma\boldsymbol{U}_\Sigma^T, \end{cases} \tag{28}$$

where $\boldsymbol{U}_R$ and $\boldsymbol{U}_\Sigma$ are orthogonal matrices, and the corresponding eigenvalue matrices are $\boldsymbol{\Lambda}_R$ and $\boldsymbol{\Lambda}_\Sigma$. The computational complexity of this eigenvalue decomposition is $\mathcal{O}(s^3n^3 + m^3)$.

By plugging (28) into (27), the correlation matrix $\boldsymbol{\Sigma}_y$ can be expressed as

$$\boldsymbol{\Sigma}_y = \boldsymbol{U}_{R\Sigma}\boldsymbol{\Lambda}_{R\Sigma}\boldsymbol{U}_{R\Sigma}^T, \tag{29}$$

where $\boldsymbol{U}_{R\Sigma} = \boldsymbol{U}_R \otimes \boldsymbol{U}_\Sigma$ and $\boldsymbol{\Lambda}_{R\Sigma} = \boldsymbol{\Lambda}_R \otimes \boldsymbol{\Lambda}_\Sigma + \delta^2\boldsymbol{I}_{smn}$. For a more detailed derivation of (29), we recommend readers to refer to the inference proposed by Stegle *et al.* (2011).

Based on (29), we can represent the inversion and the determinant of $\boldsymbol{\Sigma}_y$, that is

$$\boldsymbol{\Sigma}_y^{-1} = \boldsymbol{U}_{R\Sigma}\boldsymbol{\Lambda}_{R\Sigma}^{-1}\boldsymbol{U}_{R\Sigma}^T, \tag{30}$$

$$|\boldsymbol{\Sigma}_y| = |\boldsymbol{\Lambda}_{R\Sigma}|. \tag{31}$$

Thus, the estimate of $\boldsymbol{\beta}$ and σ^2 can be written as

$$\begin{cases} \hat{\boldsymbol{\beta}} & = \left(\boldsymbol{F}_{R\Sigma}^T\boldsymbol{\Lambda}_{R\Sigma}^{-1}\boldsymbol{F}_{R\Sigma}\right)^{-1}\boldsymbol{F}_{R\Sigma}^T\boldsymbol{\Lambda}_{R\Sigma}^{-1}\boldsymbol{Y}_{R\Sigma} \\ \hat{\sigma}^2 & = \dfrac{1}{smn}\left(\boldsymbol{Y}_{R\Sigma} - \boldsymbol{F}_{R\Sigma}\hat{\boldsymbol{\beta}}\right)^T\boldsymbol{\Lambda}_{R\Sigma}^{-1}\left(\boldsymbol{Y}_{R\Sigma} - \boldsymbol{F}_{R\Sigma}\hat{\boldsymbol{\beta}}\right), \end{cases} \tag{32}$$

where $\boldsymbol{F}_{R\Sigma} = \boldsymbol{U}_{R\Sigma}^T\boldsymbol{F} \in \mathbb{R}^{smn\times q}$ and $\boldsymbol{Y}_{R\Sigma} = \boldsymbol{U}_{R\Sigma}^T\boldsymbol{Y} \in \mathbb{R}^{smn}$. The computational complexity of (32) is $\mathcal{O}(sm^2n + s^2mn^2)$ due to the Kronecker product structure of $\boldsymbol{U}_{R\Sigma}$.

Then, the simplified log-likelihood function in (13) is derived as

$$\ell(\boldsymbol{\theta}, \delta^2) = -\frac{1}{2}\left(smn\log\hat{\sigma}^2 + \log|\boldsymbol{\Lambda}_{R\Sigma}|\right). \tag{33}$$

Since $\boldsymbol{\Lambda}_{R\Sigma}$ is an smn-order diagonal matrix, the computational complexity of (33) is $\mathcal{O}(smn)$.

With the parameters estimated by (32), the prediction of latent variables $\boldsymbol{z}^*$ in (15) can be represented as

$$\begin{cases} \hat{\boldsymbol{z}}^* & = \boldsymbol{C}_{R\Sigma}(\boldsymbol{x}^*, \boldsymbol{X})\hat{\boldsymbol{\Lambda}}_{R\Sigma}^{-1}\left(\boldsymbol{Y}_{R\Sigma} - \boldsymbol{F}_{R\Sigma}\hat{\boldsymbol{\beta}}\right) \\ \mathrm{Cov}(\hat{\boldsymbol{z}}^*) & = \hat{\sigma}^2\boldsymbol{\Sigma}_z - \hat{\sigma}^2\boldsymbol{C}_{R\Sigma}(\boldsymbol{x}^*, \boldsymbol{X})\hat{\boldsymbol{\Lambda}}_{R\Sigma}^{-1}\boldsymbol{C}_{R\Sigma}(\boldsymbol{X}, \boldsymbol{x}^*), \end{cases} \tag{34}$$

where $\hat{\boldsymbol{\Lambda}}_{R\Sigma} = \boldsymbol{\Lambda}_R \otimes \boldsymbol{\Lambda}_\Sigma + \hat{\delta}^2 \boldsymbol{I}_{smn}$ and $\boldsymbol{C}_{R\Sigma}(\boldsymbol{X},\boldsymbol{x}^*) = \boldsymbol{U}_{R\Sigma}^T \boldsymbol{C}_{zy}(\boldsymbol{X},\boldsymbol{x}^*) \in \mathbb{R}^{smn\times k}$.

The computational burden of parameter estimation and prediction is mainly due to solving the eigenvalue decomposition of both $\boldsymbol{R}_V$ and $\boldsymbol{\Sigma}_B$. Thus, the complexity of the efficient computational method for LFGP under the regular grid condition is $\mathcal{O}(s^3n^3 + m^3)$. Although this method has a lower computational complexity than that proposed in Section 4.1, it is limited by the regular grid condition and is only suitable for certain specific cases.

5. Simulation study

In this section, two synthetic functions are designed to generate data for general functional responses and closed-curve responses respectively. LFGP-IC and three functional surrogate modeling methods are selected as benchmark methods, including the blind Kriging model (BK, Joseph *et al.*, 2008), and the two-stage methods using B-spline basis (BLM, Konzen *et al.*, 2021) and functional principal components (FPCM, Chung and Kontar, 2020). We compare LFGP with these benchmark methods under both regular and irregular grid conditions to illustrate the advantages of the proposed method in terms of prediction accuracy.

Some implementation details of LFGP and the benchmark methods in this section are also provided. Specifically, in LFGP, the order of the B-spline basis and the number of control points for modeling functional responses are set to $d=3$ and $k=20$, respectively. The order of the B-spline basis and the number of control points of LFGP-IC are the same as those of LFGP, and the correlation matrix of control points is set to $\boldsymbol{\Sigma}_z = \boldsymbol{I}_k$ to satisfy the independent coefficient assumption. In BK, four input variables are selected by Bayesian variable selection from the candidate set, which includes input factors and the location variable ω, to build the Kriging model. In BLM, the order of the B-spline is $d=2$ and the number of control points is $k=15$. FPCM employs principal components accounting for the first 90% of the cumulative contribution. To evaluate the performance of these methods, we use the leave-one-out cross-validation Root Mean Square Error (RMSE) as the evaluation criterion.

5.1. Simulation for general functional responses

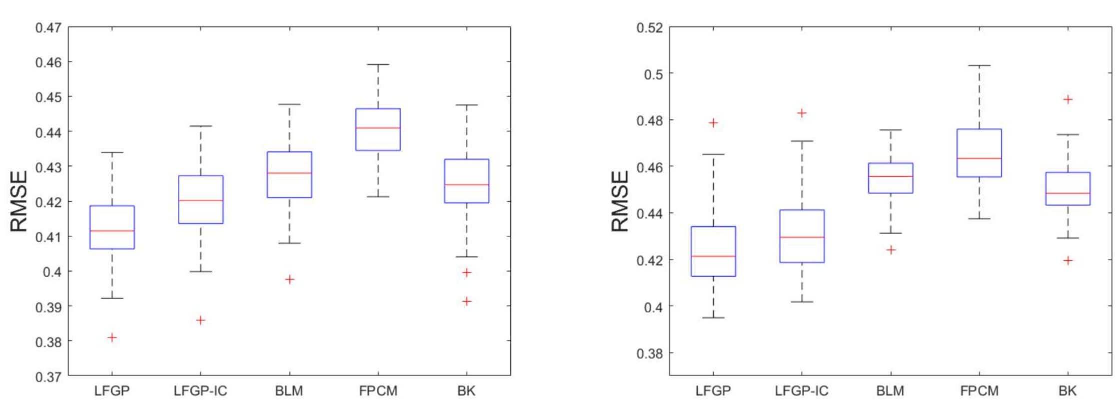
The performance of LFGP on general functional responses is studied in this subsection. The functional responses are simulated by the synthetic function from Chung and Kontar (2020), which is

$$y(\boldsymbol{x}, \omega) = 0.3\omega^2 - 2\sin\left(x_1\pi\omega^{0.85}\right) + 3\left(\arctan(\omega - 5) + \frac{\pi}{2}\right) + x_2 + \epsilon, \tag{35}$$

where $\omega \in [0, 10]$ is the location variable, $x_1 \sim \mathcal{N}(0.4, 0.03^2)$, $x_2 \sim \mathcal{U}(1.5, 6.5)$, and $\epsilon \sim \mathcal{N}(0, 0.3^2)$ denotes the measurement error. Here, $\mathcal{N}(\cdot,\cdot)$ and $\mathcal{U}(\cdot,\cdot)$ denote normal and uniform distributions, respectively.

We sample 20 inputs $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{20}\}$ in the input space using Latin hypercube sampling and generate the corresponding functional responses by (35). Two ways of collecting functional responses are designed to simulate irregular and regular grid conditions respectively. The first way collects functional responses at 20 random locations and repeats three times, while the second way collects functional responses at 20 fixed locations and repeats three times. A total of 60 functional responses are collected for each input $\boldsymbol{x}_i$, $i = 1, \ldots, 20$. By repeating this procedure 100 times, we obtain 100 sets of synthetic data under regular and irregular grid conditions, respectively. These synthetic datasets are used to evaluate the performance of LFGP, LFGP-IC, BLM, FPCM, and BK on general functional response prediction. The predictive RMSEs are shown in Figure 2.

Figure 2(a) shows the predictive RMSEs of five methods using irregular-grid functional data. According to Figure 2(a), LFGP outperforms the two-stage methods, BLM and FPCM. This result indicates that the surrogate modeling with raw functional data can utilize more functional information, and thus, achieve more accurate predictions. LFGP also shows superior performance compared with the BK model. This suggests that the nonseparable LFBS kernel



(a) RMSEs for irregular-grid functional data

(b) RMSEs for regular-grid functional data

Figure 2. Box plots of predictive RMSEs for general functional responses.

proposed in LFGP facilitates the transfer of correlation information between functional responses at different locations, improving modeling accuracy. This is mainly because the separable correlation functions used in BK induce a kind of Markov property that hinders the correlation information sharing across functional responses, leading to less accurate predictions. The comparison of LFGP and LFGP-IC shows that considering the correlations of the basis coefficients (control points) in the functional modeling can improve the prediction accuracy. This is because the output spatial correlations in LFGP are jointly represented by the spatial correlations of adjacent B-spline knots and the B-spline basis coefficients. The former is used to capture the local dependencies of functional responses, while the latter links these local dependencies to represent the global dependencies of functional responses. If the basis coefficients are assumed to be independent as in LFGP-IC, the model expressiveness would be limited, thus affecting the accuracy of the prediction.

In addition, analyzing the performance of the two-stage methods in Figure 2(a), it can be seen that BLM performs better than FPCM. This is because BLM can use the B-spline to represent the spatial correlations of neighboring functional responses, whereas FPCM with orthogonal basis functions neglects these spatial correlations, reducing the model prediction accuracy. Although BLM uses the B-spline for modeling, its performance falls short of LFGP. There are two main reasons for the failure of BLM. First, the loss of information triggered by the two-stage modeling affects the prediction accuracy of BLM. Second, unlike the correlated basis coefficients (control points) in LFGP, the B-spline basis coefficients in BLM are assumed to be independent. Therefore, BLM only uses the B-spline to represent the local dependencies of functional responses and fails to adequately capture the output spatial correlations, leading to less accurate predictions.

Figure 2(b) shows the predictive RMSEs of five methods using regular-grid functional data. The additional sparsity of the regular-grid functional data leads to an increase in the predictive RMSEs of all methods compared with Figure 2(a). Nevertheless, LFGP still outperforms the two-stage methods and the BK model. The comparison result of LFGP under regular and irregular grid conditions shows that LFGP has a higher prediction accuracy under the irregular grid condition. This is because the functional data on irregular grids is collected randomly, which can provide more functional information than the fixed collection on regular grids. If the number of output locations on regular grids is appropriately increased, the prediction accuracy of LFGP in Figure 2(b) would correspondingly improve and gradually converge to the result in Figure 2(a).

In order to compare the computational efficiency of LFGP and the benchmark methods, several experiments are provided to illustrate the prediction accuracy and computational time of these methods. Based on the synthetic dataset generated by (35), we consider 16 scenarios by varying the number of samples ($n = 15$, 20), the number of output locations ($m = 10$, 20), and the number of repetitions ($s = 2$, 3) under both irregular and regular grid conditions. The predictive RMSEs and the average runtimes under these scenarios are shown in Table 1.

Table 1. Predictive RMSEs and average runtimes of four methods in different scenarios.

Condition				LFGP		*BLM*		*FPCM*		*BK*	
Grid	*n*	*s*	*m*	*RMSE*	*time/s*	*RMSE*	*time/s*	*RMSE*	*time/s*	*RMSE*	*time/s*
Irregular	20	3	20	**0.418**	1.716	0.432	0.093	0.435	**0.044**	0.428	5.626
		2	20	**0.422**	1.226	0.443	0.086	0.464	**0.042**	0.441	2.725
		3	10	**0.519**	0.213	0.555	0.091	0.683	**0.037**	0.54	1.695
		2	10	**0.535**	0.111	0.679	0.076	0.694	**0.022**	0.664	0.805
	15	3	20	**0.449**	1.078	0.475	0.068	0.489	**0.036**	0.472	2.632
		2	20	**0.451**	0.897	0.552	0.064	0.567	**0.031**	0.514	2.343
		3	10	**0.589**	0.242	0.71	0.076	0.729	**0.028**	0.704	0.841
		2	10	**0.608**	0.142	0.719	0.075	0.746	**0.026**	0.713	0.171
Regular	20	3	20	**0.422**	1.231	0.454	0.101	0.454	**0.045**	0.437	2.097
		2	20	**0.455**	0.501	0.505	0.091	0.513	**0.048**	0.461	1.035
		3	10	**0.539**	0.192	0.617	0.103	0.698	**0.026**	0.553	0.618
		2	10	**0.552**	0.108	0.696	0.073	0.718	**0.021**	0.685	0.329
	15	3	20	**0.46**	0.891	0.505	0.06	0.531	**0.033**	0.485	1.203
		2	20	**0.505**	0.492	0.662	0.089	0.685	**0.03**	0.535	0.609
		3	10	**0.626**	0.201	0.74	0.061	0.741	**0.016**	0.73	0.339
		2	10	**0.668**	0.051	0.745	0.055	0.766	**0.015**	0.735	0.075

In accordance with the experiment results in Table 1, it can be seen that LFGP consistently provides lower predictive RMSEs compared with the other three methods in all scenarios; however, it requires more time for modeling. This suggests that considering the output spatial correlations and utilizing the original functional data in the one-stage modeling can make the prediction more accurate, although it requires more computation. Table 1 also shows that the average runtimes of the two-stage methods (BLM and FPCM) are shorter than those of LFGP and BK, and FPCM has the shortest average runtimes in all scenarios. However, the predictive RMSEs of these two-stage methods are larger than those of LFGP. The dimensionality reduction of functional data in these two-stage methods effectively reduces the modeling time, but it affects the prediction accuracy simultaneously. In all scenarios, LFGP is faster than BK, which also uses the original functional data for modeling. This result indicates that the proposed efficient computational techniques in Section 4 relax the computational complexity of LFGP, thus reducing the modeling time to some extent.

Consistent with the viewpoint proposed in Section 4.2, Table 1 shows that the average runtimes of LFGP under the regular grid condition are overall shorter than those under the irregular grid condition, indicating that LFGP has a lower computational complexity under the regular grid condition. Although LFGP has a higher complexity under the irregular grid condition, it provides lower predictive RMSEs compared with that under the regular grid condition, because the functional data collected on irregular grids can provide more information.

5.2. Simulation for closed curves

The performance of LFGP on closed-curve responses is studied in this subsection. To ensure that the endpoints of the function are smoothly connected, the closed-curve responses are simulated by a random trigonometric function

$$y(\boldsymbol{x}, \omega) = \boldsymbol{f}(\boldsymbol{x})^T \boldsymbol{\beta} + \sin(2\pi\omega)(1 + z(\boldsymbol{x})) + \epsilon, \tag{36}$$

where $\boldsymbol{f}(\boldsymbol{x})$ is a first-order polynomial for input $\boldsymbol{x} \in \mathbb{R}^{10}$, $\boldsymbol{\beta}$ represents the regression parameters, $\omega \in [0, 1]$ denotes the location variable, $z(\boldsymbol{x})$ follows a GP with square exponential kernel, and $\epsilon \sim \mathcal{N}(0, \delta^2)$ is the noise term. The parameters $\{\boldsymbol{\beta}, \boldsymbol{\theta}, \delta\}$ of this function are determined according to Wang *et al.* (2022).

We randomly sample 20 inputs $\{\boldsymbol{x}_1, ..., \boldsymbol{x}_{20}\}$ from the input datasets of Wang *et al.* (2022) and generate the corresponding functional responses by (36). Similar to Section 5.1, these functional responses are collected under both regular and irregular grid conditions. To simulate these two grid conditions, we collect functional responses at 20 random locations and 20 fixed locations and repeat this collection three times. A total of 60 functional responses are collected for each input $\boldsymbol{x}_i$, $i = 1, ..., 20$. By repeating this procedure 100 times, 100 sets of synthetic data are obtained under regular and irregular grid conditions, respectively. Using these synthetic datasets, the performance of LFGP, LFGP-IC, BLM, FPCM, and BK on closed-curve response prediction is evaluated. The corresponding predictive RMSEs are shown in Figure 3.

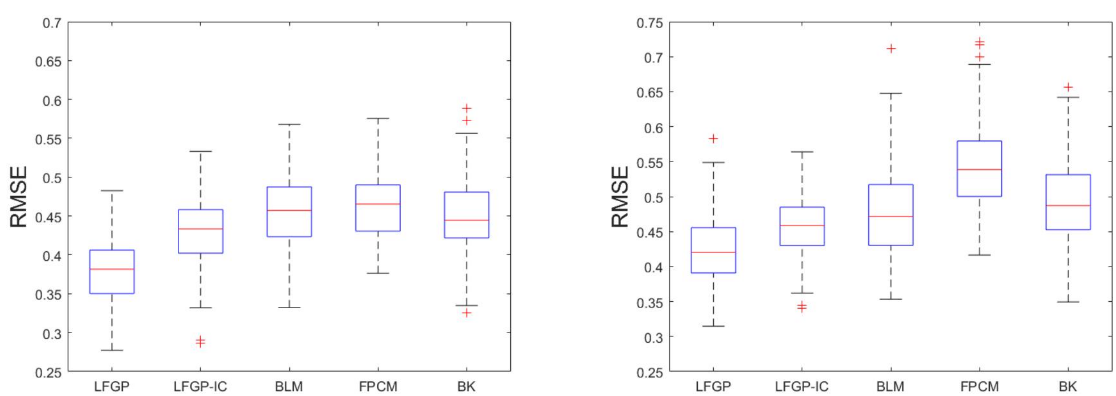
The comparison results in Figure 3 show that LFGP outperforms the two-stage methods (BLM, FPCM) and the BK model in predicting closed-curve responses under regular and irregular grid conditions. The comparison of LFGP and LFGP-IC further illustrates that considering the correlations of the basis coefficients can improve the accuracy of closed curve modeling. The superior performance of LFGP for closed-curve responses indicates that, in addition to the adequate use of functional data and the consideration of output spatial correlations, which are discussed in Section 5.1, a crucial contributing factor is the incorporation of closed shape constraints into the LFGP modeling framework. These closed shape constraints guide the parameter estimation of LFGP and thus improve the prediction accuracy for output responses at the endpoints of closed curves.

6. Case study

Composite materials have attracted considerable attention in the aircraft industry, due to their properties such as high specific modulus, high specific strength, fatigue resistance and corrosion resistance. In recent years, composite fuselage and wing structures have been widely used in the design and manufacture of large aircraft. Considering the complexity of composite mechanics, it is necessary to implement quality control during the assembly process of composite components to ensure the reliability of the assembled aircraft. In this section, we take the composite fuselage shape control of a certain aircraft as an example to study the performance of the proposed LFGP model.

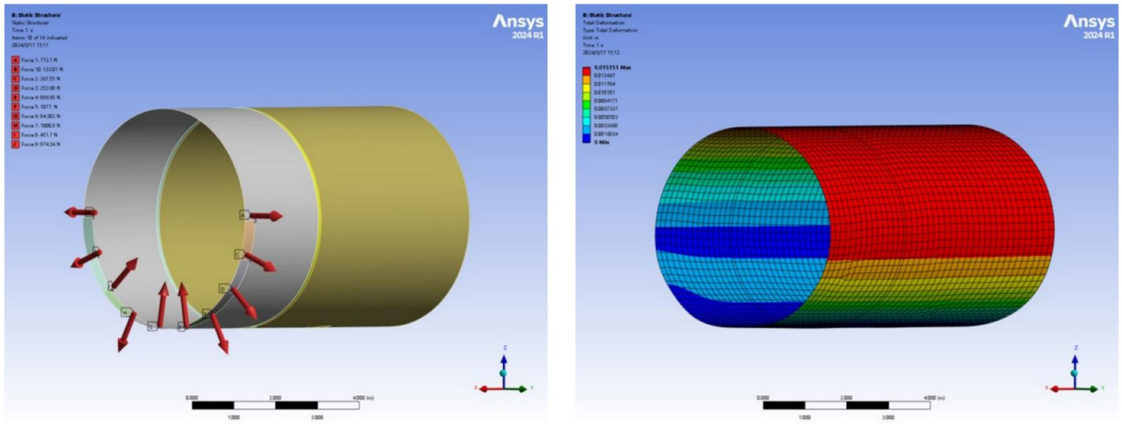
Due to different production batches and manufacturing process variations among suppliers, dimensional deformations of the composite fuselage are inevitable. To minimize the interference between the two fuselage cylinder walls during assembly, the composite fuselage needs to be adjusted to an acceptable shape in advance. The fuselage shape is adjusted by 10 actuators evenly distributed around the edge of the lower half-circle of the fuselage, each actuator providing either tension or compression.

In previous fuselage assembly processes, the forces exerted by each actuator were typically adjusted by experienced engineers. However, this tuning process is often time-consuming. In addition, the cost of using a large number of sensors to collect dimensional shape data in this process can be prohibitive. Therefore, we employ a composite fuselage simulation platform developed on the ANSYS Composite PrepPost workbench. This platform has been validated by physical experiments to accurately simulate the dimensional deformations of composite fuselages (Wen *et al.*, 2018). This platform allows us to obtain the deformation data of composite fuselages under different actuator conditions, which is used to develop surrogate models for fuselage shape control. As shown in Figure 4, the forces of 10 actuators are used as design parameters to be adapted, each ranging from -450 to 450 lbf. The output results of the simulation platform are the dimensional deformations of the composite fuselage cylinder wall in the X, Y and Z directions. Since the key to the fuselage assembly is the joint between the edges of the



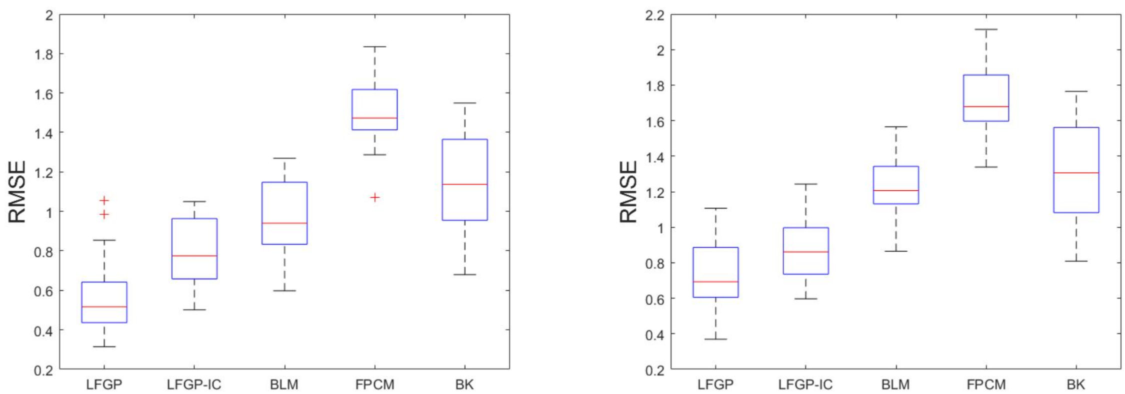
(a) RMSEs for irregular-grid functional data (b) RMSEs for regular-grid functional data

Figure 3. Box plots of predictive RMSEs for closed curve responses.



(a) Simulation of 10 actuators around fuselage (b) Simulation results of dimensional deformations

Figure 4. Composite fuselage simulation platform.



(a) RMSEs for irregular-grid deformations (b) RMSEs for regular-grid deformations

Figure 5. Box plots of predictive RMSEs for the deformations of composite fuselage.

cylinder walls, we focus only on the fuselage shape dimensions in the edge plane defined by the Y and Z coordinates.

We generate 55 combinations of actuator forces using Latin hypercube sampling. The simulation platform is then employed to simulate the dimensional deformations under these actuator forces. For each simulated cylindrical fuselage wall, we respectively select 20 random locations and 20 fixed locations along its edge in a clockwise direction and collect the fuselage dimensional deformations at the corresponding locations. This procedure is repeated three times. As the observation locations are either completely random or fixed, these collected dimensional deformations can be considered as closed-curve responses under both regular and irregular grid conditions.

Using the actuator force combination as input and the corresponding 60 sets of dimensional deformations as outputs, the functional surrogate models based on LFGP, LFGP-IC, FPCM, BLM, and BK are used to predict the dimensional deformations of the composite fuselage. With the first 30 sets of input–output combinations as training data and the remaining 25 sets as test data, we obtain the predictive RMSEs for two grid conditions in Figure 5.

Figure 5(a) shows that LFGP provides lower predictive RMSEs compared the two-stage methods (BLM, FPCM) and the BK model, indicating that LFGP has superior performance in predicting fuselage dimensional deformations. This result confirms that adequate utilization of functional information and consideration of output spatial correlations can improve the prediction accuracy. Since the basis coefficients are assumed to be independent, LFGP-IC provides larger predictive RMSEs than LFGP. Nevertheless, as a one-stage modeling method, LFGP-IC can use the original functional data for modeling and thus has better performance than the two-stage methods (BLM, FPCM). Figure 5(b) further shows that the predictive RMSEs of LFGP are larger under the regular grid condition compared with the irregular grid condition. This indicates that the regular grids only focus on the dimensional deformations at the fixed locations of the fuselage, neglecting other locations that may provide critical information. Therefore, the lack of functional information triggered by the regular grids can affect the performance of fuselage shape modeling, potentially reducing the model prediction accuracy.

Figure 6 shows the typical fitting result of LFGP to the fuselage shape, where the blue dashed line, green dashed line and red solid line represent the design values, actual values and predicted values of the composite fuselage shape, respectively. The result in Figure 6 illustrates that the predicted curve of LFGP can fit well with the actual fuselage shape. The LFGP model considering closed shape

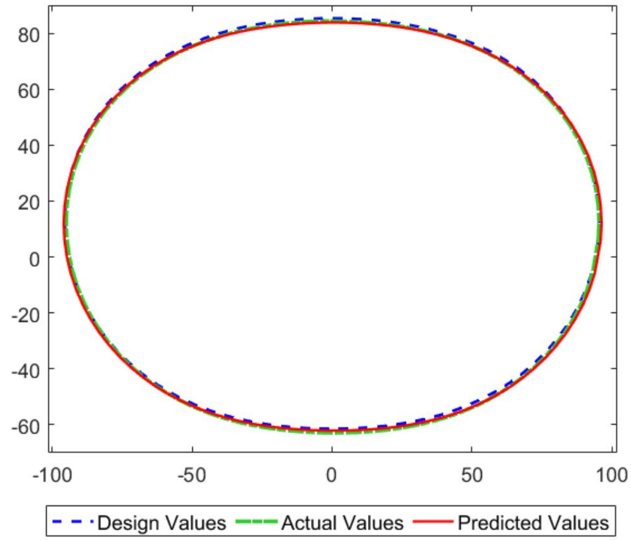


Figure 6. Typical fuselage shape fitting using LFGP.

constraints ensures the smooth connection between the endpoints of the function curve, which makes the fuselage shape modeling more reasonable and accurate. Therefore, LFGP can effectively model the fuselage shape and be further applied to the shape control in the composite fuselage assembly.

7. Conclusion

In this study, we propose the LFGP model, a novel functional surrogate modeling approach that uses B-spline basis functions to represent functional responses, where the coefficients (control points) of the B-spline are treated as random latent variables following MGP. LFGP is a one-stage modeling approach that can better utilize the raw functional data to estimate both model parameters and latent variables. The LFBS kernel of LFGP can flexibly represent the output spatial correlations at arbitrary locations on irregular grids. To accelerate this one-stage functional modeling, efficient computational techniques for LFGP under irregular and regular grid conditions are also investigated. The predictive performance of LFGP is demonstrated through some numerical examples. In addition, we extend LFGP to closed-curve surrogate modeling and apply it to predict dimensional deformations in composite fuselage shape control. Our results suggest that LFGP has significant potential to improve the accuracy of functional surrogate modeling in computer experiments.

The proposed LFGP model has potential for further refinement. First, although its formulation considers only quantitative inputs, it can be extended to the functional surrogate modeling with both quantitative and qualitative factors (Qian *et al.*, 2008), which will be explored in the future. Second, in this study, LFGP is limited to modeling two-dimensional closed curves. Our subsequent research will explore LFGP modeling for three-dimensional closed geometries (Leser *et al.*, 2017). Third, while our present study focuses primarily on functional surrogate modeling, the next phase of our research will delve into the field of functional Bayesian optimization (Maddox *et al.*, 2021) based on LFGP.

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Notes on contributors

Ziyu Liu received a BS degree in industrial engineering from Jilin University, Changchun, China, in 2020. He is currently pursuing the Ph.D. degree in industrial engineering and management at Shanghai Jiao Tong University, Shanghai, China. His current research interests include data science, uncertainty quantification, and Bayesian optimization.

Yongxiang Li received a PhD in data science from City University of Hong Kong in 2019. Currently, he is an Associate Professor in the Department of Industrial Engineering and Management at Shanghai Jiao Tong University. His research focuses on both the theoretical and applied aspects of data science integrated with domain knowledge for quality and reliability engineering using methodologies from statistics, machine learning, and signal processing. He has been working on the research areas such as computer experiments, statistical quality control, statistical anomaly detection, and intelligent fault diagnostics. He is an associate editor of *INFORMS Journal on Data Science.*

Xiaowei Yue received a BS degree in mechanical engineering from the Beijing Institute of Technology, Beijing, China, in 2011, an MS degree in power engineering and engineering thermophysics from Tsinghua University, Beijing, in 2013, and the M.S. degree in statistics and a PhD in industrial engineering with minor machine learning from the Georgia Institute of Technology, Atlanta, USA, in 2016 and 2018, respectively. He is currently an Assistant Professor in the Department of Industrial Engineering at Tsinghua University, Beijing. His research interests are focused on machine learning for advanced manufacturing. He is a Senior Member of ASQ and IISE. He serves as an Associate Editor for the *Journal of Intelligent Manufacturing* and the *IISE Transactions*.

Ershun Pan received BS and MS degrees in mechanical design and manufacturing from Northeastern University, Shenyang, China, in 1997, and a PhD in mechanical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2000. He is currently a Professor and the Head of the Department of Industrial Engineering and Management at Shanghai Jiao Tong University. His current research interests include quality control, reliability engineering and maintenance policy, production system planning and design, and lean manufacturing technology.

ORCID

Yongxiang Li http://orcid.org/0000-0003-0618-6857
Xiaowei Yue http://orcid.org/0000-0001-6019-0940

Data availability statement

The data used in the simulation study were generated using the code provided. The data for the case study were simulated using ANSYS software. The code used to generate the results can be found at https://github.com/Yongxiang-Li/LFGP.

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Appendix

Proof of Proposition 1. *We define* $\boldsymbol{B}(\boldsymbol{\omega}_i) = [\boldsymbol{b}(\omega_{i,1}), \ldots, \boldsymbol{b}(\omega_{i,m})]^T$ *for* $i = 1, \ldots, n$ *and construct a B-spline basis matrix* $\boldsymbol{B}_b = \mathrm{diag}[\boldsymbol{B}(\boldsymbol{\omega}_1), \ldots, \boldsymbol{B}(\boldsymbol{\omega}_n)] \in \mathbb{R}^{mn \times nk}$. *The correlation matrix* $\mathrm{Cor}(\boldsymbol{y}) = \{\psi_{\boldsymbol{\theta}}(y(\boldsymbol{x}, \omega), y(\boldsymbol{x}', \omega'))\}_{mn \times mn}$ *can be expressed as*

$$\mathrm{Cor}(\boldsymbol{y}) = \boldsymbol{B}_b(\boldsymbol{R} \otimes \boldsymbol{\Sigma}_{\boldsymbol{z}})\boldsymbol{B}_b^T + \delta^2 \boldsymbol{I}_{mn}.$$

Note that both $\boldsymbol{R}$ *and* $\boldsymbol{\Sigma}_{\boldsymbol{z}}$ *are positive definite, and thus we can always decompose* $\boldsymbol{R} \otimes \boldsymbol{\Sigma}_{\boldsymbol{z}}$ *into* $\boldsymbol{L}_z \boldsymbol{L}_z^T$ *via Cholesky factorization, which further implies that* $\boldsymbol{B}_b(\boldsymbol{R} \otimes \boldsymbol{\Sigma}_{\boldsymbol{z}})\boldsymbol{B}_b^T$ *is positive semi-definite. Therefore,* $\mathrm{Cor}(\boldsymbol{y})$ *is always positive definite. This completes the proof.*